



School of Informatics & IT
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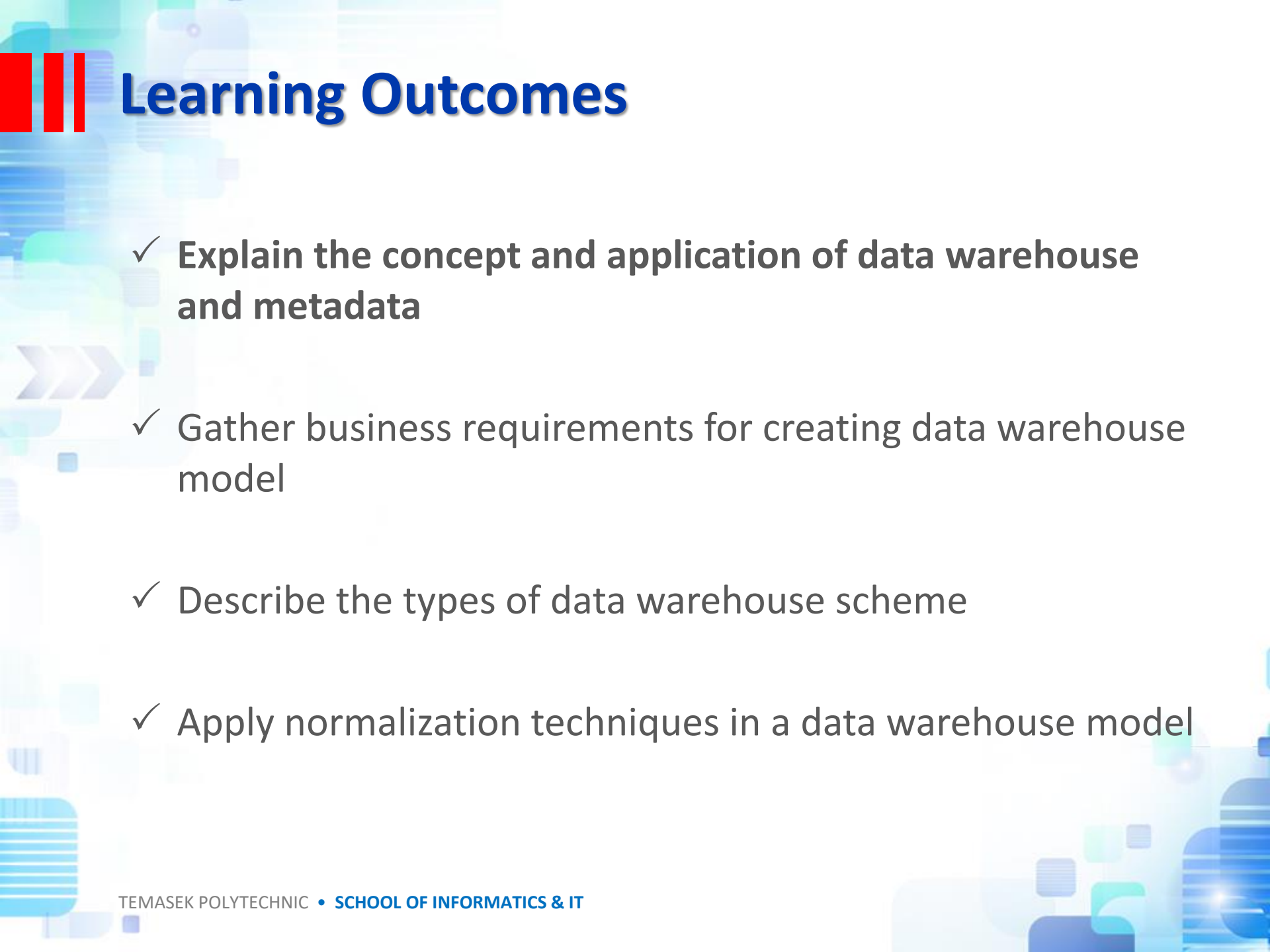
Data Warehousing and Business Intelligence

CDA2C01



Topic 2

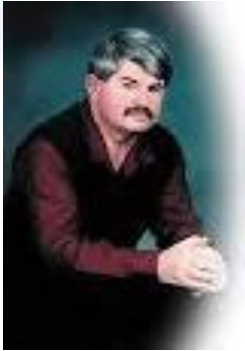
Introduction to Data Warehouse Model Design I



Learning Outcomes

- ✓ Explain the concept and application of data warehouse and metadata
- ✓ Gather business requirements for creating data warehouse model
- ✓ Describe the types of data warehouse scheme
- ✓ Apply normalization techniques in a data warehouse model

What is a data warehouse?



According to Bill Inmon, who came up with the concept of data warehousing, "a data warehouse is a subject oriented, integrated, non-volatile, and time-variant collection of data in support of management's decisions"

(Building the Data Warehouse, W.H. Inmon, John Wiley & Sons, Inc., p. 29).

The data warehouse...

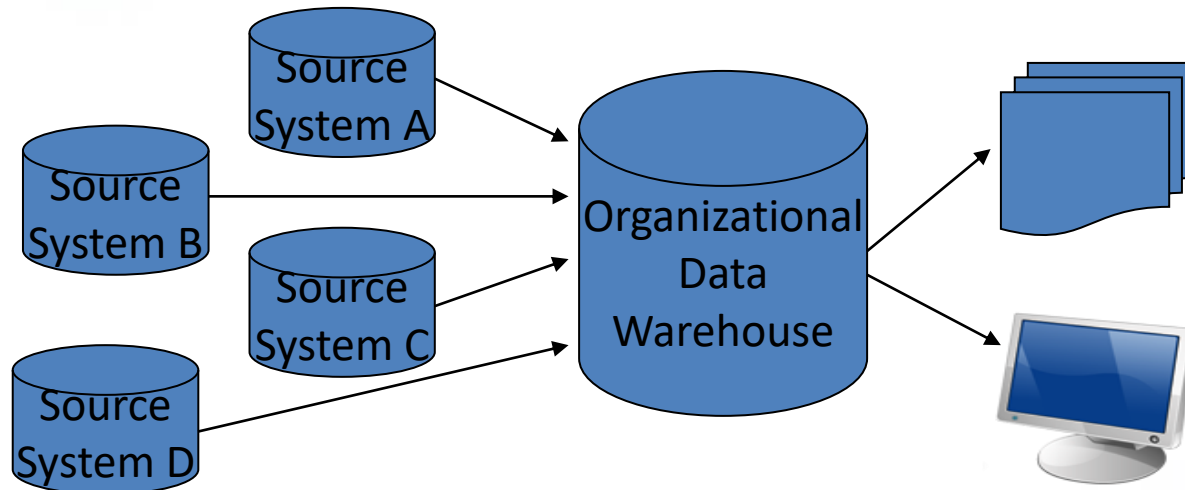
- Data warehouse usually acts as the organisation-centralized repository where consistent, detailed, summarized, current and historical data is stored.
- It contains data that changes frequently (everyday/almost everyday) transaction as well as data that does not change often non-transaction data
- It is designed for users in the organization to query, report and analyse information. This contributes to the notion of “Business Intelligence”.

Relationship between Business Intelligence and Data Warehouse

- Data Warehouse is the Infrastructure component implemented to achieve business intelligence.
- Although data warehouses provide the data to for BI, BI is **NOT** dependent on Data Warehouse to provide them with the data that they need as it can come from other sources. However data warehouse is no less important as an infrastructural component of the organization to achieve BI.

Functions of a Data Warehouse

1. **Centralised collection point** for information required for decision-making.



Typical Data Warehouse Environment

Functions of a Data Warehouse

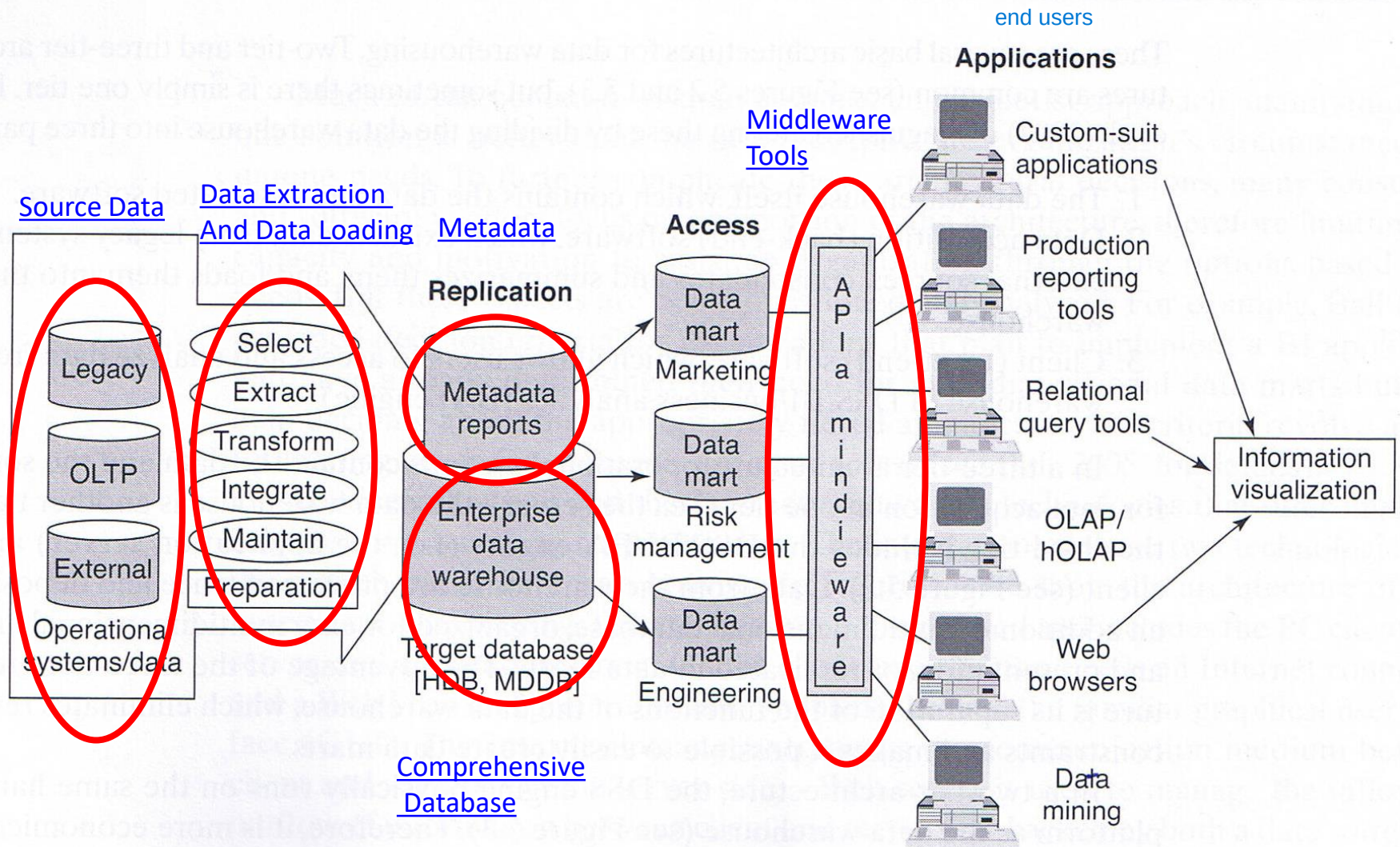
2. Direct reflection of business rules at the enterprise level

- Not limited to a specific function/business unit, it usually extends to the entire organisation.
- Requires flexibility in accommodating changes to the business rules (i.e. include new data elements, shifts in hierarchical relationships or changes to relationships between existing entities)

Functions of a Data Warehouse

3. Storage where historical data can be archived
 - To facilitate time series analysis; we can compare past and present conditions.
4. Source of information for data marts
 - Data marts are often used for exploration and data mining.
5. Source of stable data
 - The data model of the data warehouse does not change frequently. As much as possible, it is shielded from schema changes at the data source.

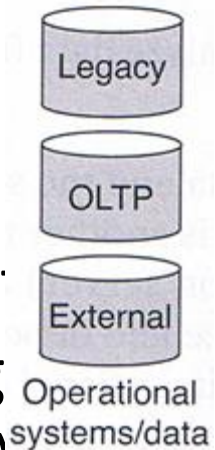
Components of Data Warehouse



Components of the Data Warehouse

1. Data Sources

- Data is usually sourced from multiple operational systems e.g. Point-of-Sales, manufacturing, reservations, service support. Also known as online transaction processing (OLTP) or enterprise resource planning (ERP) system.
- External data providers e.g. credit rating firms (S&P, Moody's) and stock market/investment data (Bloomberg, Reuters)



Components of the Data Warehouse

2. Data Extraction

- Handled by software that performs Extract Transform and Load (ETL) process



3. Data loading

- Data are loaded into the staging area, where they are transformed and cleansed. The data are then ready to load into the data warehouse.



Components of the Data Warehouse

4. Metadata

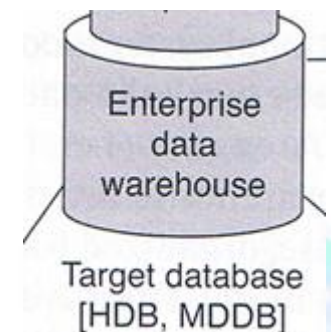
- Metadata gives meaning to the data; used for information discovery and identification. For example, metadata such as author, publisher and title helps you to search for books in a library.
- Metadata is necessary for using, building and administering data warehouse.
- Metadata glue all components together



Components of the Data Warehouse

5. Comprehensive database

- This is the Enterprise Data Warehouse (EDW) to support all decision analysis by providing relevant, summarized and detailed information originating from many different sources
- The database used can be Hierarchical Database, Relational database, Multi-Dimensional database etc.



Components of the Data Warehouse

6. Middleware tools

- Middleware tools enable end-user access to the data warehouse
- Data can be retrieved through SQL queries and used for data mining/data visualisation.



Characteristics of a Data Warehouse

Subject oriented

Data is organized around subjects (**broad** areas of interest) rather than *specific* departments/operations/applications/products (**narrow** areas of interest).

Example of subject areas in an insurance company: Customer, Product, Transaction or Activity, Policy, Claim, Account

Bill Inmon

From Wikipedia, the free encyclopedia

William H. Inmon (born 1945) is an American [computer scientist](#), recognized by many as the father of the data warehouse.^{[1][2]} Bill Inmon wrote the first book, held the first conference (with [Arnie Barnett](#)), wrote the first column in a magazine and was the first to offer classes in [data warehousing](#). Bill Inmon created the accepted definition of what a data warehouse is - a subject oriented, nonvolatile, integrated, time variant collection of data in support of management's decisions. Compared with the approach of the other pioneering architect of data warehousing, [Ralph Kimball](#), Inmon's approach is often characterized as a top-down approach.

Characteristics of a Data Warehouse

Non-Volatile

Operational databases are highly volatile (constantly changing). A query is unlikely to produce the same results if the data are frequently updated.

Unlike operational databases, data that exists in the data warehouse is usually not subject to change. Thus, same result will be produced no matter how often it is run.

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Characteristics of a Data Warehouse

Time Variant

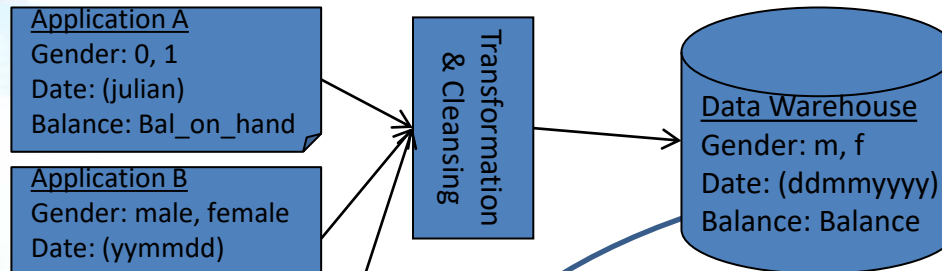
Every unit of data in the data warehouse is accurate as of some moment in time. Each record carries a date/timestamp. A time horizon is the length of time in which data exists in an environment. A 60 – 90 day time horizon is normal for operational systems; a 5 – 10 year time horizon is normal for data warehouses. Hence the data warehouse contains much more history than any other environment.

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Characteristics of a Data Warehouse



Integrated

Data warehouse obtains its data from multiple disparate sources. Hence data must be transformed into a standardised format that follows a consistent set of business rules that are understood by the organisation.

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What is Metadata?

- “What is Metadata” from YouTube by Intricity101

